

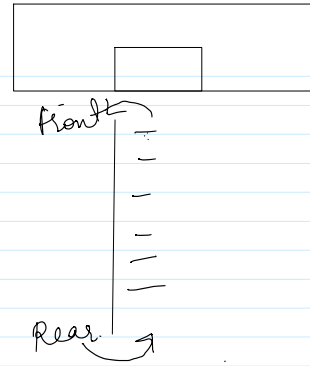
Linear Queue : Linear Data structure : utility DS ~~FIFO~~ / LIFO

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arr[5]

	[-1]	[0]	[1]	[2]	[3]	[4]
		10	150	170	20	25
Front						

Rear



```
peek()
{
    return arr[front];
}
```

Enqueue()

- ```
{
 ① Increment the rear;
 rear++;
 ② Insert the data at
 the rear position.
 arr[rear] = data;
 ③ If front == -1
 then front = 0;
```

dequeue()

```
{
 arr[front] = 0;
 front++;
}
```

Front end : pop (dequeue)

Rear end : push (enqueue)

Push (enqueue) :

- 1) Increment the rear by 1
- 2) Insert the element at the rear position
- 3) If (front == -1) front = 0;

Pop (dequeue) :

- 1) Arr[front] = 0 (optional)
- 2) Increment the front by 1

Queue full condition : rear == size-1

Queue empty condition : rear == -1 || front > rear

Always check if the queue is full before inserting the element

Always check if the queue is empty before deleting the element

